Eric Van Clepper

Contact Information
Department of Geophysical Sciences
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Education

2020 - Ph.D. Candidate, Geophysical Sciences, University of Chicago, Chicago, IL
2015 - 2019 BA *cum laude* Astronomy, Planetary Science, Northwestern University, Evanston, IL

Employment

2020 - Research and Teaching Assistant, University of Chicago, Chicago, IL
Advisor: Fred Ciesla
2019 - 2020 N'Cat Tutor, Northwestern University, Evanston, IL
2017 - 2019 Undergraduate Research Assistant, Northwestern University, Evanston, IL
Advisor: Giles Novak

Selected Grants

2023 NASA FINESST
2020 McCormick Fellowship
2018 Northwestern Weinberg Research Grant
2018 NASA ISG Summer Research Grant
2017 NASA ISG Summer Research Grant

Conference Presentations

Goldschmidt Conference, Chicago (August 2024)
FARGO3D 1st Workshop, Santiago, Chile (January 2024)
Meteoritical Society, 86th Annual Meeting, Los Angeles (August 2023)
Kavli-IAU Astrochemistry Symposium, Astrochemistry VIII, Traverse City (July, 2023)
Meteoritical Society, 84th Annual Meeting, Chicago (August 2021)

Poster Presentations

Solar System in Context, NOIRLab (October, 2025)
Origins of Life, Origin Federation Conference (Sep, 2025)
Origins of Solar Systems, Gordon Research Conference (June, 2025)
New Heights in Planet Formation, ESO Dust Busters (July, 2024)
Origins of Solar Systems, Gordon Research Conference (June, 2023)
Emerging Researchers in Exoplanet Sciences VII (August 2022)

Publications

ORCiD, Google Scholar, ADS

First Author Publications

Van Clepper, Eric, Alarcón, F., Bergin E., Ciesla F. (2025) "Dust Recycling and Icy Volatile Delivery (DRIVE): A Novel Method of Volatile Enrichment in Cold Giant Planets." *In review.*

Van Clepper, Eric, Price, E. M., and Ciesla, F. (2025) "Three-dimensional transport of solids in a protoplanetary disk containing a growing giant planet." *The Astrophysical Journal* 980:201 https://doi.org/10.3847/1538-4357/ada8a4

Van Clepper, Eric., Bergner, J., Bosman A., Bergin E., and Ciesla F. (2022) "Chemical Feedbacks of Pebble Growth: Impacts on CO Depletion and C/O Ratios." *The Astrophysical Journal* 927:206. https://doi.org/10.3847/1538-4357/ac511b

Other Publications

Price, Ellen M., **Van Clepper, E.**, and Ciesla F. (2025) "Dynamics of Small, Constant-size Particles in a Protoplanetary Disk with an Embedded Protoplanet." *The Astrophysical Journal* 979:37 https://doi.org/10.3847/1538-4357/ad9f35

Seligman, Darryl Z., Rogers, L., Cabot, S., et al. **incl. Eric Van Clepper** (2022) "The Volatile Carbon-to-oxygen Ratio as a Tracer for the Formation Locations of Interstellar Comets." *The Planetary Science Journal* 3:150. https://doi.org/10.3847/PSJ/ac75b5.

Chuss, David T., Andersson, B-G., Bally, J., et al. **incl. Eric Van Camp*** (2019) "HAWC+/SOFIA Multiwavelength Polarimetric Observations of OMC-1." *The Astrophysical Journal* 872:187. https://doi.org/10.3847/1538-4357/aafd37.

Santos, Fabio P., Chuss, D., Dowell C., et al. **incl. Eric Van Camp*** (2019) "The Far-Infrared Polarization Spectrum of ρ Ophiuchi A from HAWC+/SOFIA Observations." *The Astrophysical Journal* 882: 113. https://doi.org/10.3847/1538-4357/ab3407.

**Name change from Eric Van Camp to Eric Van Clepper in 2021*

Software Development

`CANDY` https://github.com/ervc/newcandy

Chemistry ANd DYnamics, coupled astrochemistry and dust evolution for self-consistent modeling of chemical composition during planetesimal formation.

`ParTrace` https://github.com/ervc/cpartrace

Particle Tracking alogrithm to track trajectories of tracer particles in 3D Hydrodynamic disk simulations, accounting for gas drag with 3D flows around embedded planets.

Academic Service

Departmental Service

2024	Lead TA for intro to coding Bootcamp for incoming students
2024 - 2025	Graduate Union Steward for the Department of Geophysical Sciences
2023	Co-organizer of University of Chicago Graduate Research Expo
2022 - 2023	Co-organizer of University of Chicago Exoplanet Journal Club

Outreach

2025, July 23	Guest Lecture on planet formation for UChicago summer class ASTR11901
2023, Nov 14	Planetary science and grad student life panel, College of the South
2022, Nov 21	Planetary science and grad student life panel, College of the South